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Separation of electrostatic and magnetic phase shifts using a modified transport-of-intensity equation

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ABSTRACT

We introduce a new approach for the separation of the electrostatic and magnetic components of the electron wave phase shift, based on the transport-of-intensity equation (TIE) formalism. We derive two separate TIE-like equations, one for each of the phase shift components. We use experimental results on FeCoB and Permalloy patterned islands to illustrate how the magnetic and electrostatic longitudinal derivatives can be computed. The main advantage of this new approach is the fact that the differences in the power spectra of the two phase components (electrostatic phase shifts often have significant power in the higher frequencies) can be accommodated by the selection of two different Tikhonov regularization parameters for the two phase reconstructions. The extra computational demands of the method are more than compensated by the improved phase reconstruction results.

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1. Introduction

The ability to image the magnetic domain structure in magnetic materials is critical to understanding the way in which these materials respond to an external field and how this behavior is related to the microstructure of the material. Applications of magnetic materials are wide-ranging and include for example magnetic recording media in which the magnetic domains control bit size, and permanent magnets. There are a number of techniques that can be used to image the magnetic domain structure [1] of which the highest spatial resolution is obtainable using Lorentz transmission electron microscopy (LTEM) [2]. This technique has the additional advantage that the microstructure and composition of the materials can be explored in the same instrument for direct correlation with the domain structure. Recent examples of the use of Lorentz microscopy are the study of nanoscale magnetic interactions in square artificial spin ice lattices, in particular the structure of Dirac strings and "magnetic monopoles" [3,4], *in situ* studies of the motion of domain walls in Ni_2MnGa type ferromagnetic shape memory alloys [5] and the interaction of magnetic domain walls with twin boundaries in FePdCo alloys [6].

In LTEM one measures the phase shift of the incident electron beam due to the sample's magnetic induction, which leads to a deflection of the electron beam. This is achieved by defocusing the imaging lens by a large amount [7], a technique known as Fresnel imaging. Typical deflection angles are in the range of tens to hundreds of microradians, i.e., about two orders of magnitude smaller than typical Bragg angles, so that defocus values of several to hundreds of micrometers are not unusual. The large defocus generates significant defocus blurring, so that the spatial resolution of Lorentz images is usually estimated to be around 10 nm. Along with the defocus blurring, in field emission instruments the Fresnel images also suffer from significant delocalization due to the large spherical aberration of the long focal length Lorentz lens used for this type of observation; C_s values of several meters are not unusual [8].

The combination of a Lorentz pole piece in which the sample can be imaged whilst sitting in a low magnetic field, unlike in conventional TEM objective lenses, with an imaging spherical-aberration corrector provides improved spatial resolution in Lorentz mode, because the correction of C_s down to a few microns lowers the amount of delocalization, and, hence, the amount of defocus needed to render the magnetic domain walls visible. The reduced defocus, in turn, leads to reduced defocus blurring and therefore an improved spatial resolution of the order of 1 nm [9–11]. In addition, the reduced defocus brings the Fresnel image mode into the realm of applicability (the so-called *small defocus limit* [12]) of the transport-of-intensity equation (TIE) formalism for electron wave phase reconstruction [13,14].

The TIE formalism can be used to reconstruct the phase of the electron wave at the detector plane from a through-focus series centered at zero defocus. Due to the small angles involved in Lorentz deflection, one can show that the application of TIE to Lorentz images enables the phase shift of the sample exit wave function to be obtained, provided the small defocus limit is

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observed. The total phase shift consists of two contributions: an electrostatic phase shift due to the mean inner potential and, potentially, any localized electrostatic polarization, and a magnetic phase shift due to the magnetic induction of the sample. In most cases, the magnetic phase shift is rather weak compared to the electrostatic component, so that it can become difficult to extract the magnetic information. It is then common to record a second through-focus series with the sample flipped upside down, since the lack of time-reversal symmetry of the magnetic phase shift will change its sign relative to the electrostatic phase shift. A simple subtraction then produces the magnetic phase shift.

In this contribution, we present an alternative approach to the separation of electrostatic and magnetic phase shifts using the TIE formalism. In Section 2 we describe the transport-of-intensity equation and how it is usually solved using fast Fourier Transforms (FFTs). Then we introduce a new approach for phase shift separation, based on the linearity of the TIE formalism, and illustrate its benefits using a computational model. Finally, in Section 3 we present a series of experimental phase reconstructions on a patterned Permalloy sample and on amorphous rectangular CoFeB islands. We conclude with a brief summary of results.

2. Theory

2.1. The transport-of-intensity equation

Under paraxial conditions, the phase of an electron wave is coupled to the intensity gradient along the electron trajectory. Paganin and Nugent [13] derived the so-called transport-of-intensity equation (TIE) formalism, which expresses this relation mathematically. If the total phase shift of the electron wave is represented by the symbol $\varphi_t = \varphi_e + \varphi_m$, with φ_e the electrostatic and φ_m the magnetic phase shifts, then the TIE reads

$$\nabla \cdot (I_0 \nabla \varphi_t) = -k\partial_z I, \tag{1}$$

where I_0 is the intensity of the in-focus image, $k = 2\pi/\lambda$ is the wave number, and the z-direction is parallel to the electron propagation direction. The ∇ differential operators are two-dimensional and operate in the plane normal to the z-axis, and the symbol ∂_z represents the partial derivative with respect to the variable z. It should be noted that the gradient of the magnetic component of the total phase shift can be related to the integrated magnetic induction by the relation [15]

$$\nabla \varphi_m = -\frac{\pi}{\phi_0}(\mathbf{B} \times \hat{\mathbf{n}})t, \tag{2}$$

where t is the foil thickness, $\hat{\mathbf{n}}$ a unit vector along the beam direction, and ϕ_0 the flux quantum. Note that this relation is only meaningful for samples with uniform thickness and vanishing fringing fields.

The TIE formalism can be derived in a number of different ways, notably from a paraxial approximation to the free-space Schrödinger equation [16], or from the real-space propagator equation [17]. For Lorentz microscopy conditions, with small scattering angles, the TIE follows directly from a small-angle approximation to the Lorentz lens point-spread function [18]. An elegant numerical solution approach to the TIE formalism was proposed by Paganin and Nugent [13]; their approach uses Fourier transforms to implement the inverse differential operators ∇^{-1} and ∇^{-2}. The total phase shift can be written formally by the following expressions:

$$\phi_t = -k\nabla^{-2}\left[\nabla \cdot \left(\frac{1}{I_0}\nabla(\nabla^{-2}[\partial_z I_0])\right)\right]; \tag{3}$$

$$\phi_t = -k\nabla^{-1} \cdot \left(\frac{1}{I_0}(\nabla^{-1}[\partial_z I_0])\right). \tag{4}$$

Comparing this relation with Eq. (2) indicates that the integrated magnetic induction components can be obtained by partially solving the TIE formalism using a single application of the inverse gradient operator ∇^{-1} [19, p. 655], provided that there are no spatial variations in the electrostatic phase shift. However, this is unlikely to be the case for realistic LTEM samples as there will be electrostatic potential variations due to either sample thickness variations or to the lack of materials in between patterned nanostructures. For a derivation of the conditions under which the TIE formalism is applicable to Lorentz images we refer to Ref. [12]. A description of the boundary conditions used to solve the equation can be found in [20].

2.2. Separation of electrostatic and magnetic phase shifts

In this section, we describe a new approach to the separation of electrostatic and magnetic phase components. In the conventional approach, which we will summarize first, one reconstructs the total phase shift, $\varphi_t^+ = \varphi_e + \varphi_m$, using a through-focus series, I_-^+, I_0^+, and I_+^+, for the sample in the upright orientation, indicated by the superscript $+$; the defocus amount is indicated by the subscript. The TIE in this case is given by

$$\nabla \cdot (I_0^+ \nabla \varphi_t^+) = -k\partial_z I^+ \approx -\frac{k}{2\,\Delta f}(I_+^+ - I_-^+). \tag{5}$$

The second equality is valid in the small defocus limit and approximates the longitudinal derivative as the difference between overfocus and underfocus images.

When the sample is flipped upside down, a second through-focus series I_-^-, I_0^-, and I_+^- is used to solve a second TIE for the total phase $\varphi_t^- = \varphi_e - \varphi_m$ (the sign change reflects the fact that the magnetic phase shift is not invariant with respect to time reversal symmetry):

$$\nabla \cdot (I_0^- \nabla \varphi_t^-) = -k\partial_z I^- \approx -\frac{k}{2\Delta f}(I_+^- - I_-^-). \tag{6}$$

The electrostatic and magnetic phase shifts are then obtained by adding and subtracting the two reconstructed phases:

$$\varphi_e = \frac{1}{2}[\varphi_t^+ + \varphi_t^-]; \tag{7}$$

$$\varphi_m = \frac{1}{2}[\varphi_t^+ - \varphi_t^-]. \tag{8}$$

This traditional approach can be used with the TIE formalism as well as with standard off-axis holography phase reconstructions.

The linearity of the transport-of-intensity equation suggests that separation of the phase shifts may be accomplished also by substituting the expressions for φ_t^+ and φ_t^- into the respective TIEs (5) and (6):

$$\nabla \cdot (I_0 \nabla \varphi_e) + \nabla \cdot (I_0 \nabla \varphi_m) = -k\partial_z I^+;$$
$$\nabla \cdot (I_0 \nabla \varphi_e) - \nabla \cdot (I_0 \nabla \varphi_m) = -k\partial_z I^-,$$

where we have made use of the fact that $I_0^+ = I_0^- \equiv I_0$. Adding and subtracting these equations we obtain

$$\nabla \cdot (I_0 \nabla \varphi_e) = -k\partial_z^e I \quad \text{[TIE–e]} \tag{9}$$

$$\nabla \cdot (I_0 \nabla \varphi_m) = -k\partial_z^m I \quad \text{[TIE–m]} \tag{10}$$

where we define

$$\partial_z^e I \equiv \frac{1}{2}[\partial_z I^+ + \partial_z I^-]; \tag{11}$$

$$\partial_z^m I \equiv \frac{1}{2}[\partial_z I^+ - \partial_z I^-], \tag{12}$$

as the electrostatic and magnetic longitudinal intensity derivatives, respectively. In practice, these derivatives are determined from two pairs of underfocus/overfocus images in which case the longitudinal derivatives can be written as

$$\partial_z^e I = \frac{1}{4\,\Delta f}(I_+^+ - I_-^+ + I_+^- - I_-^-); \tag{13}$$

$$\partial_z^m I = \frac{1}{4\,\Delta f}(I_+^+ - I_-^+ - I_+^- + I_-^-). \tag{14}$$

If a longer through-focus series is available, then the longitudinal derivatives can be computed by fitting a polynomial to the entire through-focus series for each image pixel; details about this fitting procedure are discussed in Section 2.3. Once the electrostatic and/or magnetic longitudinal derivatives are known, then the equations representing TIE-e and TIE-m can be solved *independently* for the corresponding phase shifts.

In addition to separating the electrostatic and magnetic phase shifts from the start, the new phase reconstruction approach also allows for a modification of the traditional Tikhonov regularization that is applied to the algorithm for the case of the inverse Laplacian ∇^{-2}. To compute the inverse Laplacian by means of Fourier transforms, one modifies the Fourier space kernel as follows:

$$\frac{1}{q^2} \rightarrow \frac{q^2}{(q^2+q_c^2)^2}, \tag{15}$$

where q_c is a low frequency, typically in the range of $q_c = 0.01-0.2\ \mathrm{nm}^{-1}$. This approach stabilizes the phase reconstruction against noise in the low frequency components. It is likely that the power spectra of the electrostatic and magnetic components are quite different; magnetic phase shifts are typically relatively smooth and slowly varying, whereas the electrostatic phase shift, especially in the presence of interfaces or the sample's edge, will show significant power in the higher spatial frequencies. This is illustrated in Fig. 1, where the simulated electrostatic and magnetic phase shifts along with the corresponding power spectra are shown for a set of rectangular Permalloy islands. It is clear that the electrostatic phase shift has contributions from higher spatial frequencies as compared to the magnetic phase shift. Since, in our new approach, the separation of electrostatic and magnetic contributions occurs *before* the TIE is solved, one can now employ different Tikhonov regularization parameters q_c for the two reconstructions, which should improve both electrostatic and magnetic phase reconstructions.

2.3. Determination of the electrostatic and magnetic longitudinal derivatives

The longitudinal intensity derivatives are typically determined using the central difference method as described in Eqs. (5) and (6). If more than three through-focus series images are recorded, the central difference method can be expanded to include more terms and reduce the error. However, a more accurate method for determining the longitudinal derivative can be implemented by fitting a polynomial to the entire through-focus series for each image pixel [21]. For each pixel (i, j) of the aligned image set, a quadratic polynomial can be fitted through the intensity values:

$$I_{ij}(z) = \sum_{k=0}^{2} A_k(i,j) z^k. \tag{16}$$

Interpreting the polynomial as a truncated Taylor series expansion with respect to the defocus z at $z=0$, the factors $A_k(i,j)$ represent the derivatives of the image intensity at each pixel. Hence, an image formed with the values $A_1(i,j)$ corresponds to the longitudinal derivatives. A least squares fit can be performed using the singular value decomposition (SVD) method to determine the parameters $A_k(i,j)$. In the experimental images, the standard deviation values at each image pixel (which are needed for the

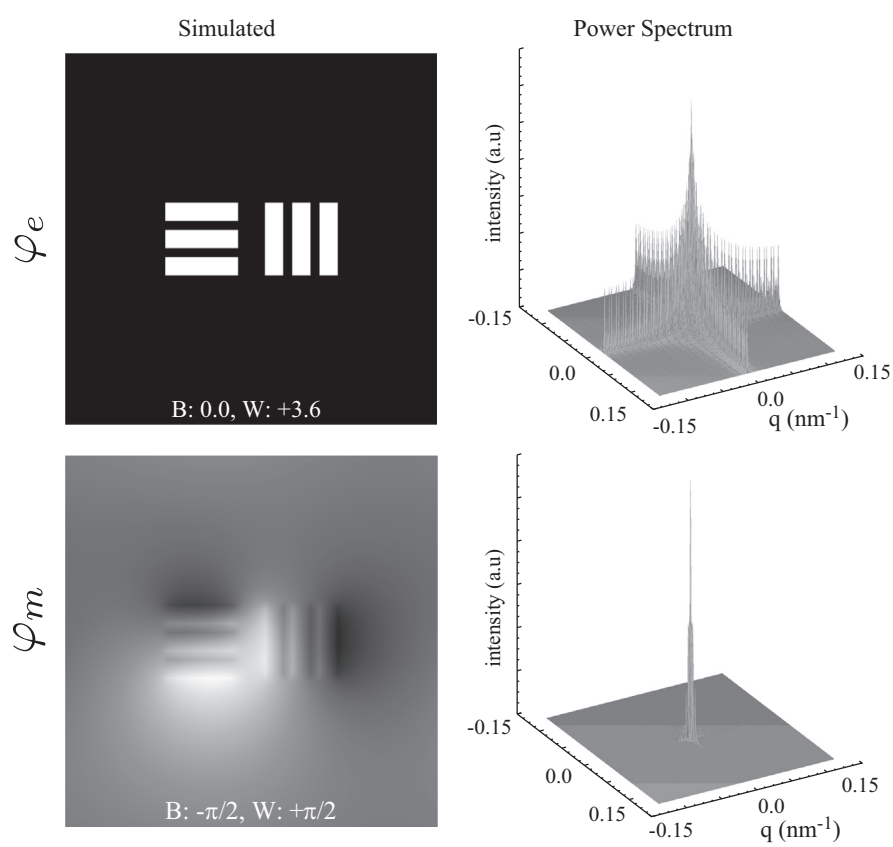

Fig. 1. Simulated electrostatic and magnetic phase shifts for a set of 6 rectangular Permalloy islands and the corresponding power spectrum for each phase shift.

SVD approach) were determined from a 16-pixel neighborhood around each pixel.

In order to solve both the TIE-e and TIE-m equations, we need to determine the separate longitudinal derivatives $\partial_z^e I$ and $\partial_z^m I$. The fitting procedure described above can be extended to determine the magnetic and electrostatic longitudinal derivatives by directly incorporating time reversal symmetry into the fitting equations. For each image pixel, we write the entire through-focus series of upright and flipped image intensities as a single vector as follows:

$$[I^+_{-N\Delta z}, \ldots, I^+_0, \ldots, I^+_{N\Delta z}, I^-_{-N\Delta z}, \ldots, I^-_0, \ldots, I^-_{N\Delta z}],$$

where Δz is the defocus step in the through-focus series consisting of $2N+1$ images. With this vector we can associate a list of defocus values as follows:

$$[[-N\Delta z, \ldots, 0, \ldots, N\Delta z]^+, [-N\Delta z, \ldots, 0, \ldots, N\Delta z]^-]$$

If we carry out the SVD least squares fitting on these vectors, then the result directly produces the electrostatic longitudinal derivative $\partial_z^e I$. To determine the magnetic longitudinal derivative, $\partial_z^m I$, we simply reverse the sign of the defocus values of the second half of the defocus list

$$[[-N\Delta z, \ldots, 0, \ldots, N\Delta z]^+, [N\Delta z, \ldots, 0, \ldots, -N\Delta z]^-]$$

to reflect the absence of time reversal symmetry for the magnetic phase shift, and then carry out the same fitting procedure. Using this modified method of polynomial fitting, the magnetic and electrostatic longitudinal derivatives can be determined separately and directly from the entire through-focus series.

3. Experimental methods

The modified phase reconstruction algorithm was used to reconstruct the magnetic and electrostatic phase shifts on two sets of samples:

1. A 100 nm thick patterned film of amorphous CoFeB was deposited on a TEM grid with a SiO_x membrane using DC magnetron sputtering. Additional seed and capping layers of Ta (2 nm) were used to protect the thin film. The film was then patterned, using a bitmap patterning method, into rectangular islands $5.6 \times 1\ \mu m^2$ in size using an FEI Nova 600 NanoLab focused ion beam instrument, operated at 30 kV.
2. An array of Permalloy islands of 20 nm thickness and various aspect ratios was fabricated using electron beam lithography directly on SiN TEM window grids.

The patterned samples were inserted directly, without exposure to external magnetic fields, into an aberration-corrected JEOL 2100F Lorentz transmission electron microscope operating at 200 kV. This microscope has a dedicated Lorentz pole piece where the sample sits in a magnetic field-free region and is equipped with an imaging spherical aberration corrector capable of reducing C_s from 120 mm to less than 100 μm. In-focus bright field images of both samples are shown in Fig. 2. For each sample, two through-focus series with 25 μm step size were obtained, one with the sample in the standard upright orientation, the other with the sample flipped upside down. For the CoFeB sample, each through-focus series consisted of 3 images, whereas for the permalloy sample 9 images were recorded for each series, centered symmetrically around the in-focus image.

4. Experimental results and analysis

4.1. Longitudinal intensity derivatives

Before phase reconstructions can be attempted, all images in each of the datasets must be registered with respect to each other with sub-pixel accuracy. While the defocus step size of 25 μm is relatively small compared to that used for traditional Lorentz images—the image aberration corrector allows for the use of significantly smaller defocus values—the act of defocusing the images introduces several artifacts: a small image shift proportional to the defocus, and a small magnification change across the image series. All images were aligned with respect to the in-focus images using a hybrid algorithm of cross correlation (for coarse alignment) and maximization of the mutual information (for sub-pixel alignment) [22]. For the images corresponding to the upside-down sample orientation, a mirror operation is applied to the images, and the rotation angle between the upright and upside-down image sets is determined as part of the registration procedure. Fig. 3 shows six images from the CoFeB sample superimposed on each other after alignment: the edges of the individual images can be seen around the perimeter of the superposition image. No image rotations were detected as a function of defocus, and all images were rotated using bicubic interpolation to orient the islands horizontally. A similar alignment approach was employed for the Permalloy dataset.

For the CoFeB phase reconstructions, the electrostatic and magnetic longitudinal derivatives were approximated using Eqs. (13) and (14), and are shown in Fig. 4(a) and (b), respectively.

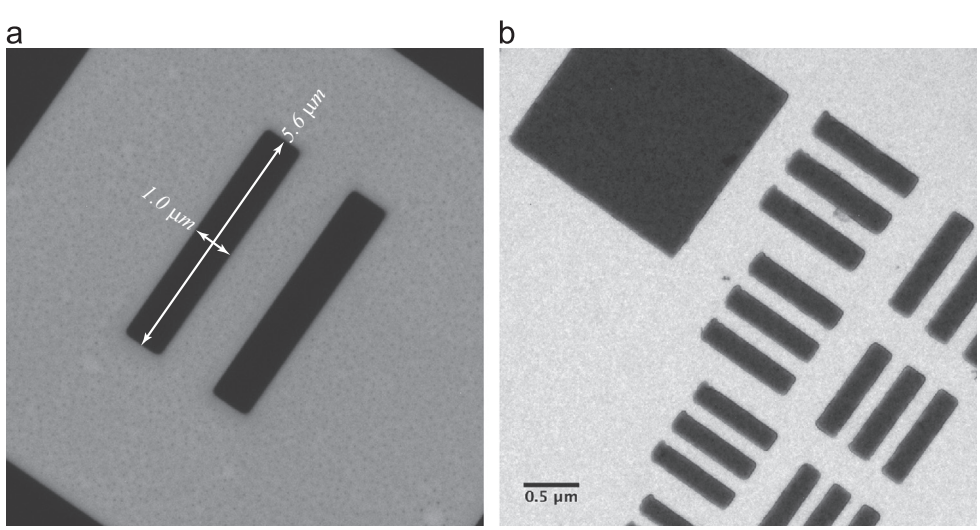

Fig. 2. In-focus images of (a) a pair of amorphous CoFeB islands and (b) a patterned Permalloy film.

Note that the small particles on the support grid (debris from the ion milling step) are sharply defined, an indication that the image registration is accurate, since small misalignments would give rise to strong contrast features in these difference images. In the electrostatic image one can clearly observe a thin FIB damage layer around the perimeter; this layer may also be present on top of the islands.

For the Permalloy patterned sample, two different sets of longitudinal derivatives were computed, one by using only three images from each 9-image through-focus series (central difference method for a defocus value of $\pm 100\,\mu m$), and one by using the modified polynomial fitting method to the complete through-focus series.

Fig. 5 shows a comparison between the two-image central difference derivatives (a and c) and the derivatives determined via the modified polynomial fitting procedure (b and d). For both the electrostatic and the magnetic derivative, the modified fitting procedure gives rise to images that are less noisy than the central difference procedure. Fig. 5(e) shows the intensity histogram for the central difference image (blue line), and the modified fitting image (red line). From the histogram it is clear that there is an improved signal-to-noise ratio indicated by a sharper peak. This is also indicated by comparing the variance in the images. The variance in the image obtained using the central difference approach is one order of magnitude higher than that obtained using the polynomial fitting method.

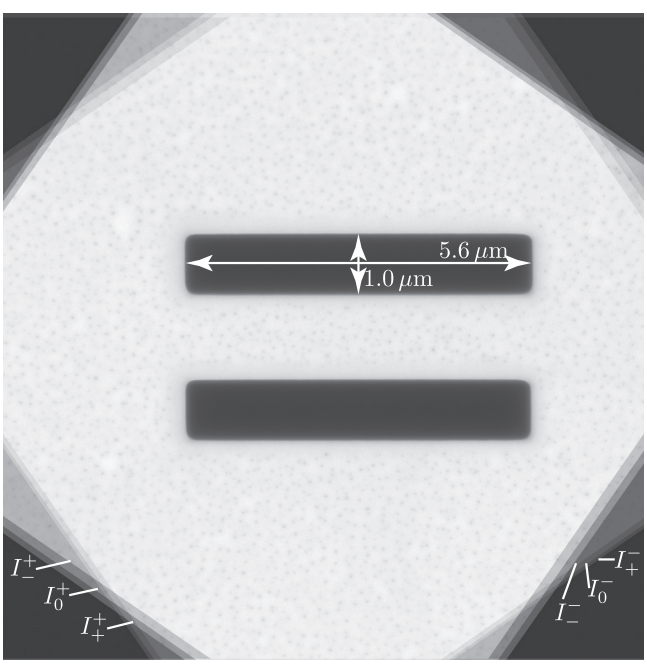

Fig. 3. Superposition of six CoFeB images after registration; the two groups of three images correspond to the through-focus series for the upright (+) and flipped (−) sample orientations, indicated by the superscripts on the image labels.

4.2. Phase reconstructions

4.2.1. CoFeB

Phase reconstructions were carried out by solving the electrostatic and magnetic transport-of-intensity equations (9) and (10) using the inverse Laplacian FFT approach with Tikhonov regularization. Fig. 6 shows the electrostatic phase shift (top row) and the B_y integrated induction component derived using Eq. (2) (bottom row) for three different values of the Tikhonov regularization parameter q_c. The first column displays the results without regularization; the electrostatic phase shift shows the typical slow intensity variations caused by noise in the low frequency components of the longitudinal derivative $\partial_z^e I$. The magnetic induction component B_y, being a gradient component of the magnetic phase shift, is less sensitive to slowly varying intensity changes in the magnetic phase shift.

For a Tikhonov parameter of q_c=0.001, some of the low frequency variations in the electrostatic phase shift have disappeared but the phase shift still shows significant signs of blurring. The B_y magnetic induction component, on the other hand, shows well-defined magnetic domains across the island and a flat background in the support film region as expected. For a regularization parameter of q_c=0.01, the electrostatic phase shift no longer shows the effect of low frequency noise and displays a uniform background as well as a clearly defined surface damage region and island interior; note that the phase shift drops off from the island center towards the surface layer, in agreement with the fact that the top island surfaces are slightly rounded and hence the islands are somewhat thicker in the middle than near the edges. The magnetic induction component B_y for q_c=0.01 on the other hand is poorly defined and rather noisy; this is due to the fact that the larger regularization parameter has also suppressed some of the low frequency magnetic information in the phase shift, which propagates to a loss of domain visibility in the integrated induction map.

It is important to realize that in the traditional regularized TIE reconstruction, only a single value of q_c can be used, since it is the total phase shift φ_t that is reconstructed, not the individual phase shifts φ_e and φ_m. Hence, our new separation method for electrostatic and magnetic phase shifts has a distinct advantage over the traditional approach.

4.2.2. Permalloy

Phase reconstructions for the Permalloy sample were carried out using the electrostatic and magnetic transport-of-intensity equations (9) and (10) and the inverse Laplacian FFT approach. Since a longer through-focus series was recorded from this sample, a comparison can be made of the reconstructed phases for the central difference method and the modified SVD fitting

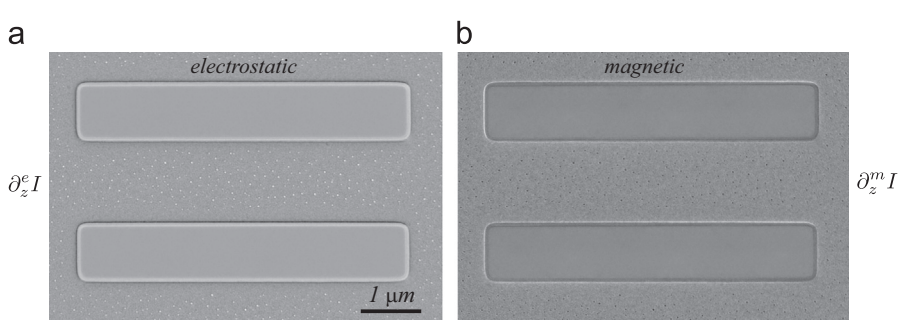

Fig. 4. (a) Electrostatic and (b) magnetic longitudinal derivatives for the CoFeB dataset, computed using Eqs. (13) and (14).

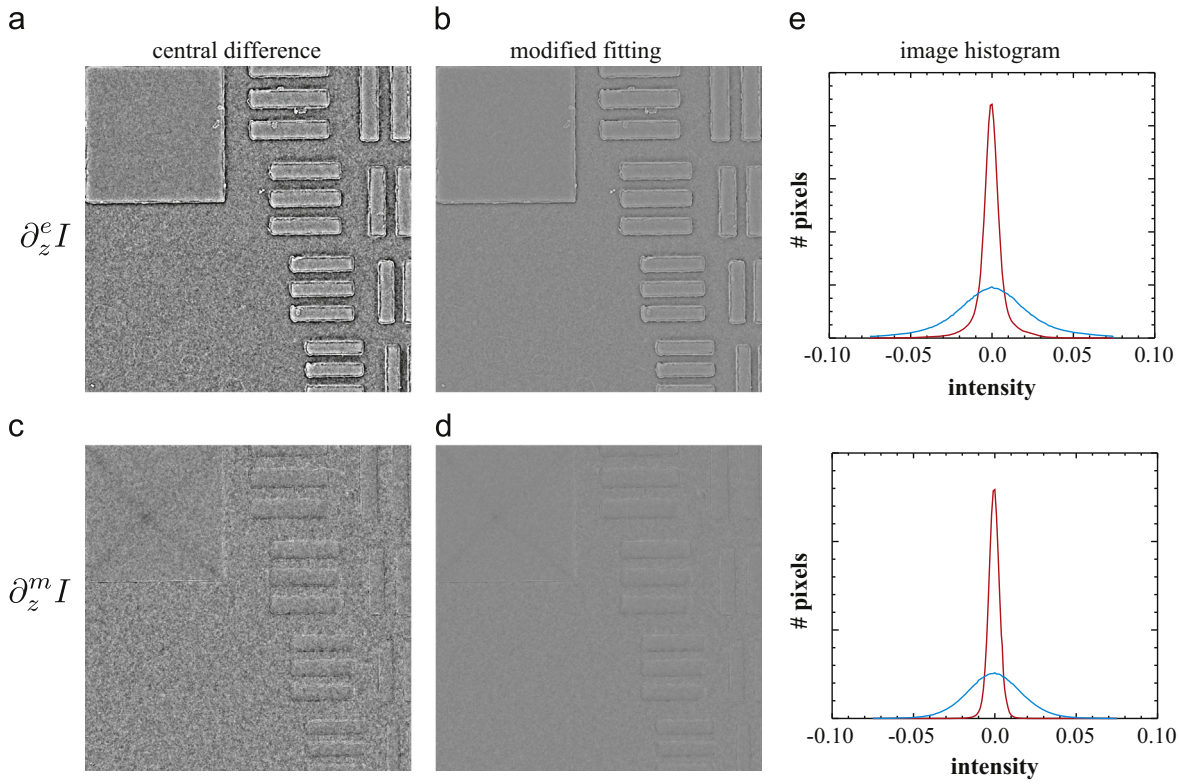

Fig. 5. Electrostatic (top row) and magnetic (bottom row) longitudinal derivatives for the Permalloy dataset, using the two-image central difference method (a), (c); and the modified polynomial fitting method (b), (d). (e) shows the intensity histogram (with the intensity in arbitrary units with respect to the mean intensity) of $\partial_z^e I$ and $\partial_z^m I$ for the central difference method (blue) and for the modified fitting method (red). (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

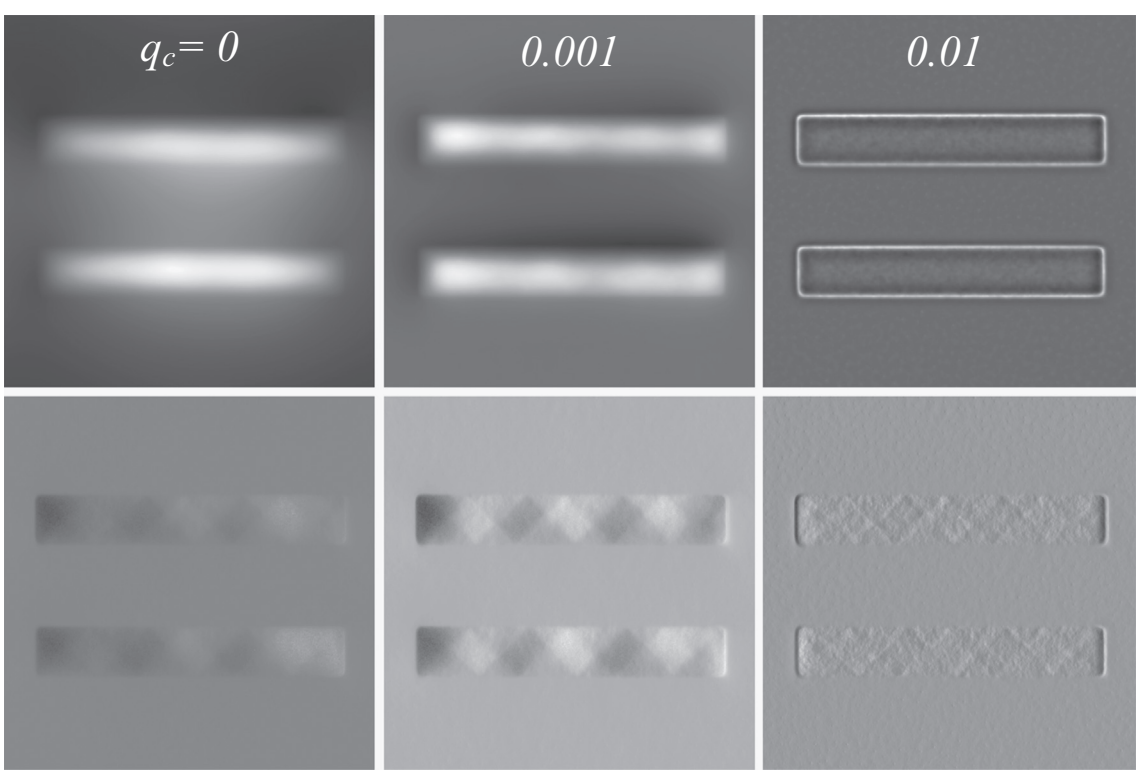

Fig. 6. Electrostatic phase shift (top row) and B_y component of the integrated magnetic induction (bottom row) for the CoFeB sample as a function of the Tikhonov regularization parameter q_c.

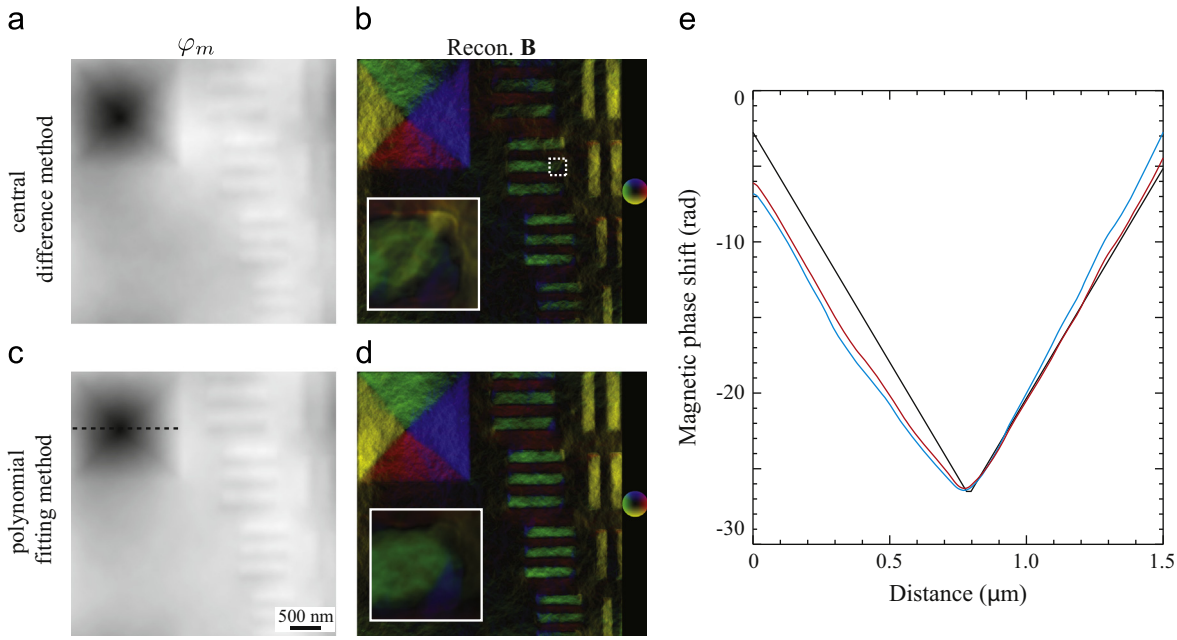

Fig. 7. Phase reconstructions for the Permalloy sample: (a) and (c) are the reconstructed magnetic phase shifts and (b), (d) the corresponding magnetic induction maps for the central difference approach (top row) and the modified fitting method (bottom row). (e) is a lone plot along the dashed line in (c) comparing the theoretical (black), central difference (blue) and modified fitting (red) results. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

method. Fig. 7 shows the reconstructed magnetic phase shift (a, c) and the magnetic induction map (b,d) obtained from the central difference method (top row) and the modified fitting method (bottom row). Since the signal-to-noise ratio of the derivative determined by the SVD fitting method is significantly larger than that by the central difference method (see Fig. 5) the resulting magnetic induction map shows a similar improvement. The domain boundaries are well defined in the square structure in the upper left corner. The 'S' or 'C' domain states in the rectangular islands are also better reconstructed by the fitting method. This can be seen more clearly in the insets in (b) and (d), which show close-ups of the highlighted regions. In these reconstructions, no Tikhonov regularization was used. Fig. 7(e) shows a line plot along the dashed line in (c) comparing the reconstructed phase shifts from central difference method (blue), polynomial fitting method (red) and theoretical phase shift (black).

5. Summary

In this paper, we have introduced a new approach to separating the electrostatic and magnetic components of the electron wave phase shift, based on the transport-of-intensity equation formalism. Making use of the linearity of the TIE and the principle of time reversal symmetry, we have derived two separate TIE-like equations, one for each of the phase shift components. This requires a redefinition of the longitudinal derivative that goes in the right-hand side of the equation. We have illustrated how the magnetic and electrostatic longitudinal derivatives can be computed for the case of a three-image through-focus series, where the derivative is essentially a central difference, and a longer through-focus series, where the derivative in each image pixel is computed from a parabolic fit to the intensity as a function of defocus. The main advantage of this new approach is the fact that the differences in the power spectra of the two phase components (electrostatic phase shifts often have significant power in the higher frequencies) can be accommodated by the selection of two different Tikhonov regularization parameters for the two phase reconstructions. The extra computational demands of the method are more than compensated by the improved phase reconstruction results.

An interesting question, that has so far remained unanswered, is whether or not the selection of the appropriate Tikhonov parameters can be automated. As is clearly seen from the reconstructions in Fig. 6, the effect of changing q_c is somewhat similar to a blurring of the resulting image; perhaps one can adapt autofocusing algorithms used in cameras to the automatic determination of the optimal regularization parameter. In addition, the regularization itself affects the magnitude of the reconstructed phase shift in that the total power in the phase shift spectrum is reduced with increasing q_c. Both of these questions form the topics of ongoing research.

Acknowledgments

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